

Image Metadata (EXIF) Analysis Using ExifTool

Practice case report | ExifTool | Case003_ImageMetadata

1. Case Details

Examiner	Lucky Bhati
Analysis date	05 October 2026
Case name	Case003_ImageMetadata
Purpose	Practice analysis of two image files: a photo taken by the examiner and a camera image recovered in Case001

2. Objective

To extract and compare the embedded metadata (EXIF) of a modern smartphone photo and an older digital camera image, to check whether either file stores GPS location data, and to compare embedded timestamps with file system timestamps.

3. Tools Used

- ExifTool 13.59 (metadata extraction)
- Windows PowerShell (Get-FileHash) for evidence hashing

4. Evidence Details

File 1	IMG_20261005_131834.jpg: photo taken by the examiner with a smartphone (5.8 MB)
File 1 SHA-256	35E6B00BD689A6E555D002594A735D786CD4F10C8C50BE1157EC5DA94C449FA8
File 2	recovered_f0000000.jpg: image recovered by file carving in Case001 from a Canon camera memory card image (856 kB)
File 2 SHA-256	FDD665A0845745D84391D1A6527C550E0E038D79F64D9DAA8B62EBB92937E8A0
Location used	D:\Forensics\Project 3

5. Methodology

- Calculated the SHA-256 hash of both files before analysis.
- Ran **exiftool.exe** on each file and reviewed the full output.
- Ran **exiftool.exe -gps:all** on the smartphone photo to check for location data.
- Compared embedded (EXIF) dates with the file system dates shown by ExifTool.
- Recorded the findings with screenshots.

6. Findings

6.1 Device and capture details. The two files come from very different devices (Figures 1 to 4):

Field	File 1 (smartphone)	File 2 (recovered camera image)
Make	Xiaomi	Canon
Model	Redmi Note 9 Pro Max	Canon PowerShot SD800 IS
Date/Time Original	2026-10-05 13:18:35.030	2008-12-23 14:12:38
Image size	4640 x 3472 (16.1 MP)	1600 x 1200 (1.9 MP)
Aperture / Shutter / ISO	f/1.9, 1/100 s, ISO 198	f/2.8, 1/80 s, ISO 75

Field	File 1 (smartphone)	File 2 (recovered camera image)
Focal length	5.4 mm (25 mm equivalent)	4.6 mm (27.8 mm equivalent)
Flash	Off, did not fire	Off, did not fire
Other	Rear sensor, HDR off, orientation Rotate 90 CW	Canon firmware 1.00, file number 100-0001, quality Superfine, owner name blank

6.2 GPS check. The command **exiftool.exe -gps:all** returned no output for the smartphone photo, which means no GPS tags are stored in that file (Figure 2). No GPS fields were seen in the reviewed output of the Canon image either.

6.3 Embedded dates versus file system dates.

Date type	File 1 (smartphone)	File 2 (recovered camera image)
Embedded (EXIF) date	2026-10-05 13:18:35	2008-12-23 14:12:38
File modification	2026-10-05 13:18:36 +05:30	2026-10-02 09:59:00 +05:30
File creation	2026-10-05 13:26:56 +05:30	2026-10-03 21:33:12 +05:30
File access	2026-10-05 13:30:58 +05:30	2026-10-05 13:31:00 +05:30

For File 2, the file system dates (October 2026) show when the file was extracted and copied on the examiner's computer, while the EXIF date (December 2008) shows when the photo was actually taken. For File 1, the file creation time (13:26:56) is later than the EXIF capture time (13:18:35), which suggests the file was copied to this computer after it was taken. File system dates describe the copy of the file, not the original event, so the embedded EXIF date is the more useful value.

6.4 Unique identifier. The Canon image carries an Image Unique ID (c76d7a8c83d35d21d3188aa1627153e2) and the camera's file number (100-0001). Such identifiers can help link an image to a specific camera or card (Figure 5).

7. Conclusion

The metadata identified the make, model, settings and capture date of both images. The smartphone photo contained no GPS location data. The comparison showed that file system dates change when a file is copied or extracted, whereas the embedded EXIF capture date stays with the image. This is why examiners record both and rely on the hash and EXIF data to describe the evidence.

8. Limitations

- Only two files were analysed.
- The reason there is no GPS data in the smartphone photo (for example the camera location setting) was not verified.
- The GPS-only command was run on the smartphone photo, not on the Canon image.
- EXIF data can be edited or removed, so it should be supported by other evidence.
- The email header part of the planned metadata work is not included in this report.

9. Chain of Custody Note

Date	Action	By
05 Oct 2026	Copied the two files to D:\Forensics\Project 3 and calculated SHA-256 hashes	Lucky Bhati
05 Oct 2026	Ran ExifTool (read-only analysis, files not modified) and took screenshots	Lucky Bhati

10. Evidence Screenshots

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS D:\Forensics\Project 3> .\exiftool.exe IMG_20261005_131834.jpg
ExifTool Version Number      : 13.59
File Name                    : IMG_20261005_131834.jpg
Directory                   : .
File Size                    : 5.8 MB
File Modification Date/Time  : 2026:10:05 13:18:36+05:30
File Access Date/Time       : 2026:10:05 13:30:58+05:30
File Creation Date/Time     : 2026:10:05 13:26:56+05:30
File Permissions             : -rw-rw-rw-
File Type                   : JPEG
File Type Extension         : jpg
MIME Type                   : image/jpeg
Exif Byte Order              : Big-endian (Motorola, MM)
Camera Model Name            : Redmi Note 9 Pro Max
Orientation                  : Rotate 90 CW
Modify Date                  : 2026:10:05 13:18:35
Y Cb Cr Positioning         : Centered
ISO                          : 198
Exposure Program             : Program AE
F Number                     : 1.9
Exposure Time                : 1/100
Mirror                       : false
Sensor Type                  : rear
Hdr                           : off
Op Mode                      : 36869
Small Picture                : false
AI Scene                     : 0
Filter Id                    : 66048
Zoom Multiple                : 1
Sensing Method               : Not defined
```

Figure 1: ExifTool output for the smartphone photo (device, date, settings)

```
Windows PowerShell
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PS D:\Forensics\Project 3> .\exiftool.exe -gps:all IMG_20261005_131834.jpg
PS D:\Forensics\Project 3> |
```

Figure 2: GPS check on the smartphone photo: no GPS tags returned

```
Windows PowerShell
PS D:\Forensics\Project 3> .\exiftool.exe recovered_f0000000.jpg
ExifTool Version Number      : 13.59
File Name                    : recovered_f0000000.jpg
Directory                   : .
File Size                    : 856 kB
File Modification Date/Time   : 2026:10:02 09:59:00+05:30
File Access Date/Time        : 2026:10:05 13:31:00+05:30
File Creation Date/Time       : 2026:10:03 21:33:12+05:30
File Permissions              : -rw-rw-rw-
File Type                    : JPEG
File Type Extension           : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Little-endian (Intel, II)
Make                         : Canon
Camera Model Name             : Canon PowerShot SD800 IS
Orientation                  : Horizontal (normal)
X Resolution                  : 180
Y Resolution                  : 180
Resolution Unit               : inches
Modify Date                   : 2008:12:23 14:12:38
Y Cb Cr Positioning           : Centered
Exposure Time                 : 1/80
F Number                      : 2.8
Exif Version                  : 0220
Date/Time Original            : 2008:12:23 14:12:38
Create Date                   : 2008:12:23 14:12:38
Components Configuration      : Y, Cb, Cr, -
Compressed Bits Per Pixel     : 5
Shutter Speed Value           : 1/79
Aperture Value                : 2.8
Max Aperture Value            : 2.8
Flash                        : Off, Did not fire
Focal Length                  : 4.6 mm
```

Figure 3: ExifTool output for the recovered Canon image (make, model, dates)

```
Windows PowerShell
Focal Type                    : Zoom
Focal Plane X Size            : 5.84 mm
Focal Plane Y Size            : 4.39 mm
Auto ISO                      : 75
Base ISO                      : 100
Measured EV                   : 10.91
Target Aperture               : 2.8
Target Exposure Time          : 1/79
Exposure Compensation         : 0
White Balance                 : Auto
Slow Shutter                  : Off
Shot Number In Continuous Burst : 0
Optical Zoom Code             : 0
Flash Guide Number            : 0
Flash Exposure Compensation    : 0
Auto Exposure Bracketing      : Off
AEB Bracket Value             : 0
Control Mode                  : Camera Local Control
Focus Distance Upper          : 0.38 m
Focus Distance Lower          : 0 m
Bulb Duration                 : 0
Camera Type                   : Compact
Auto Rotate                   : None
ND Filter                     : Off
Self Timer 2                  : 0
Flash Output                  : 0
Canon Image Type              : IMG:PowerShot SD800 IS JPEG
Canon Firmware Version        : Firmware Version 1.00
File Number                   : 100-0001
Owner Name                    :
ISO                           : 75
Rotation                      : 0
Camera Temperature            : 24 C
```

Figure 4: Canon image: firmware, file number and camera details

```
Windows PowerShell
Canon Model ID      : PowerShot SD800 IS / Digital IXUS 850 IS / IXY Digital 900 IS
AF Area Mode       : Auto
Num AF Points      : 9
Valid AF Points     : 9
Canon Image Width   : 1600
Canon Image Height  : 1200
AF Image Width      : 1536
AF Image Height     : 230
AF Area Widths      : 276 276 276 276 276 276 276 276
AF Area Heights     : 41 41 41 41 41 41 41 41
AF Area X Positions : -276 0 276 -276 0 276 -276 0 276
AF Area Y Positions : -41 -41 -41 0 0 0 41 41
AF Points In Focus  : 0
Primary AF Point    : 0
Thumbnail Image Valid Area : 0 0 0 0
Date Stamp Mode     : Off
My Color Mode       : Off
Firmware Revision   : 1.00 rev 5.00
Categories          : (none)
Face Detect Frame Size : 1600 1200
Face Width          : 35
Faces Detected      : 0
Image Unique ID     : c76d7a8c83d35d21d3188aa1627153e2
User Comment        :
Flashpix Version    : 0100
Color Space         : sRGB
Exif Image Width    : 1600
Exif Image Height   : 1200
Interoperability Index : R98 - DCF basic file (sRGB)
Interoperability Version : 0100
Related Image Width  : 1600
Related Image Height : 1200
Focal Plane X Resolution : 7111.111111
```

Figure 5: Canon image: Image Unique ID